

Content Review Workflows -- Migration Analysis (Major Review & Simple Review)

1. Executive Summary

1.1 Purpose

The site uses two content review workflow variants -- **Major Review** and **Simple Review** -- that share the same custom handler codebase, the same infrastructure (replication, email, scheduling), and the same publication pipeline. They differ only in the number of approval gates before content is published.

Variant	Approval Gates	Use Case
Major Review	3 gates: Design/UX → Editorial → Final Production	Content requiring multi-team review
Simple Review	1 gate: Final Production only	Content requiring only Web Ops approval

Both workflows share the same rework loop, delayed release mechanism, email notification system, cancellation handling, and publication steps. The Simple Review is a **subset** of the Major Review -- it is the same workflow with the Design/UX and Editorial review gates removed.

1.2 Migration Impact -- At a Glance

The business logic of both workflows is **fully preserved** in AEM Cloud Service. No approval gates are removed, no participant roles change, and no business rules are altered.

The changes are entirely at the **infrastructure level** and apply equally to both variants:

- **How content is published:** Sling Content Distribution replaces agent-based replication
- **How emails are sent:** Cloud-compatible email service .
- **How cache is invalidated:** EDS CDN handles automatically (Dispatcher flush steps removed)
- **How custom code is registered:** OSGi Declarative Services annotations replace deprecated Felix SCR annotations
- **How workflow routes:** OOTB Goto Step replaces custom jump handler for unconditional routing

The workflow experience for **content authors, reviewers, and web operations teams remains the same** for both variants.

All custom handlers are shared between both workflows. Refactoring a handler once applies to both variants.

1.3 Key Characteristics

Characteristic	Major Review	Simple Review
Review Stages	3 (Design/UX, Editorial, Final Production)	1 (Final Production only)
Rework Loop	Reject at Design/UX → rework	Reject at Final Production → rework
Scheduled Release	Yes -- reviewer sets activation date	Yes -- same mechanism
Force Deploy	Yes -- bypass scheduled wait	Yes -- same mechanism
Email Notifications	8 notification points	5 notification points
Cancellation	From any review stage	From any review stage
Team Routing	Web Ops team selected by author at start	Same
Total Steps (Current)	43	29

1.4 References

Note: Both workflows were analyzed on QA. Steps documented are as per the workflows on QA. Translation was confirmed to not be part of these workflows (in discovery session) and is not considered in this analysis.

Major Review QA Reference: <http://tfaem-author-qa1-use1-0.aemqa.thermofisher.net:4502/conf/global/settings/workflow/models/lifetech/lifetech-major-review-publication.html>

Simple Review QA Reference: <http://tfaem-author-qa1-use1-0.aemqa.thermofisher.net:4502/conf/global/settings/workflow/models/lifetech/lifetech-simple-review-publication.html>

2. Current Workflow Definitions (AEM 6.4)

2.1 Major Review -- Complete Step Sequence (43 steps)

```
1  1. Log Start Workflow                                [LogWorkflowHandler -
auto-advance]
2  2. Set Rework Parameters (steppedBack=false)
[PlaceParametersHandler - auto-advance]
3  3. Select Web Operations Team                        [Dialog Participant
Step -> lt-wf-author]
4  4. Place Web Ops Team on Workflow Metadata
[DialogParameterHandler - auto-advance]
5  5. Jump Over Rework if First Loop
[JumpToNodeIfSetHandler -> "Design/UX Review"]
6  6. Log Rework Content                                [LogWorkflowHandler -
auto-advance]
7  7. Send Rework Notification Email
[EmailNotificationsHandler - auto-advance]
8  8. Rework Content Step                              [Dynamic Participant -
> ECMA initiator-chooser]
9  9. OR Split: Done with Rework / Cancel
10  --- Branch 1: Log "Done" -> continue
11  --- Branch 2: Log "Cancel" -> Jump to Cancel Processing
12 10. Set steppedBack=true
[PlaceParametersHandler - auto-advance]
13 11. Report Metadata                                [LogWorkflowHandler -
auto-advance]
14 12. Replicate to Preview
[ReplicateToAgentHandler -> push-to-preview]
15 13. Cache Flush Preview
[ReplicateToAgentHandler -> flush-preview]
16 14. Notify Design Reviewer Email
[EmailNotificationsHandler - auto-advance]
17 15. Review: UX / Design                            [Participant Step ->
lt-wf-design-reviewer]
18 16. OR Split: Approve / Reject / Cancel
19  --- Branch 1: Log "Approve" -> continue
20  --- Branch 2: Log "Reject" -> Jump back to step 6 (Rework)
21  --- Branch 3: Log "Cancel" -> Jump to Cancel Processing
22 17. Log Start Editorial Review                      [LogWorkflowHandler -
auto-advance]
23 18. Replicate to Preview
[ReplicateToAgentHandler -> push-to-preview]
24 19. Cache Flush Preview
[ReplicateToAgentHandler -> flush-preview]
25 20. Notify Editorial Reviewer Email
[EmailNotificationsHandler - auto-advance]
26 21. Review: Editorial                              [Participant Step ->
lt-wf-editorial-reviewer]
27 22. OR Split: Approve / Cancel
28  --- Branch 1: Log "Approve" -> continue
29  --- Branch 2: Log "Cancel" -> Jump to Cancel Processing
30 23. Log Start Final Production Review                [LogWorkflowHandler -
auto-advance]
31 24. Replicate to Preview
[ReplicateToAgentHandler -> push-to-preview]
32 25. Cache Flush Preview
[ReplicateToAgentHandler -> flush-preview]
33 26. Notify Final Production Reviewer Email
[EmailNotificationsHandler - auto-advance]
34 27. Review: Final Production                        [Dynamic Participant -
> PropertiesParticipantChooser]
35  (Dialog: Absolute Timer Delay for delayed release date selection)
36 28. OR Split: Approve / Cancel
37  --- Branch 1: Log "Approve" + Place Delayed Release Date ->
continue to step 29
```

```

38     +-- Branch 2: Log "Cancel" -> Jump to Cancel Processing
39 29. Log Start Production Run [LogWorkflowHandler -
    auto-advance]
40 30. Replicate to Preview
    [ReplicateToAgentHandler -> push-to-preview]
41 31. Cache Flush Preview
    [ReplicateToAgentHandler -> flush-preview]
42 32. Check Delayed Release Date
    [CheckDelayedReleaseDateHandler - auto-advance]
43 33. Report Delayed Release as Comment
    [SetDelayedReleaseAsCommentHandler - auto-advance]
44 34. Send Pending Publication Email
    [EmailNotificationsHandler - auto-advance]
45 35. Waiting to Publish [Dynamic Participant -
    > AbsoluteTimeAutoAdvancer timeout]
46 36. OR Split: Force Deploy / Cancel
47     +-- Branch 1: Log "Force Deploy" -> continue
48     +-- Branch 2: Log "Cancel" -> Jump to Cancel Processing
49 37. Replicate to Production
    [ReplicateToAgentHandler -> push-to-production]
50 38. Cache Flush Production
    [ReplicateToAgentHandler -> flush-production]
51 39. Send Finished Publication Email
    [EmailNotificationsHandler - auto-advance]
52 40. Jump to End (skip cancel processing) [JumpToNodeHandler -
    auto-advance]
53 41. Cancel Workflow Processing [LogWorkflowHandler -
    auto-advance]
54 42. Inform Content Author about Cancellation
    [EmailNotificationsHandler - auto-advance]
55 43. Log End of Workflow [LogWorkflowHandler -
    auto-advance]
56

```

2.2 Simple Review -- Complete Step Sequence (29 steps)

The Simple Review workflow uses the same handlers as Major Review. Steps marked with (=Major) are identical to the corresponding Major Review steps. Differences are marked with **[DIFFERS]**.

```

1 1. Log Start Workflow [LogWorkflowHandler -
    auto-advance] (=Major Step 1)
2 2. Set Rework Parameters (steppedBack=false)
    [PlaceParametersHandler - auto-advance] (=Major Step 2)
3 3. Select Web Operations Team [Dialog Participant
    Step -> lt-wf-author] (=Major Step 3)
4 4. Place Web Ops Team on Workflow Metadata
    [DialogParameterHandler - auto-advance] (=Major Step 4)
5 5. Jump Over Rework if First Loop
    [JumpToNodeIfSetHandler -> "Final Production"] (=Major Step 5)
6 **[DIFFERS]** Target:
    "Start the Final Production Review"
7 (Major targets
    "Design/UX Review")
8 6. Rework Content (Email Notification)
    [EmailNotificationsHandler - auto-advance] (=Major Step 7)
9 **[DIFFERS]**
    Template: ../simple/review/rejected/en.txt
10 (Major uses
    ../major/review/rejected/en.txt)
11 7. Rework Content Step [Dynamic Participant -
    > ECMA initiator-chooser] (=Major Step 8)
12 8. OR Split: Done / Cancel (=Major Step 9)

```

```

13      +-- Branch 1: Log "Done" -> continue
14      +-- Branch 2: Log "Cancel" -> Jump to Cancel Processing
15      (NO Design/UX Review -- steps 10-16 of Major are ABSENT)
16      (NO Editorial Review -- steps 17-22 of Major are ABSENT)
17      (NO Set steppedBack=true -- Major Step 10 is ABSENT)
18      **[DIFFERS]**
19      9. Log Start Final Production Review          [LogWorkflowHandler -
auto-advance]          (=Major Step 23)
20      10. Replicate to Preview
[ReplicateToAgentHandler -> push-to-preview]          (=Major Step 24)
21      11. Cache Flush Preview
[ReplicateToAgentHandler -> flush-preview]          (=Major Step 25)
22      12. Notify Final Production Reviewer
[EmailNotificationsHandler - auto-advance]          (=Major Step 26)
23      **[DIFFERS]**
Template: .../simple/review/webops/en.txt
24      Notification group:
lt-wf-simple-notification (Major: lt-wf-major-notification)
25      13. Review: Final Production          [Dynamic Participant -
> PropertiesParticipantChooser] (=Major Step 27)
26      (Dialog: Absolute Timer Delay for delayed release date selection)
27      14. OR Split: Approve / Cancel / Reject
**[DIFFERS]**
28      +-- Branch 1: Approve + Place Delayed Release Date -> continue
29      +-- Branch 2: Cancel -> Jump to Cancel Processing
30      +-- Branch 3: **Reject -> Jump back to Rework Content**
**[DIFFERS]**
31      (Major does NOT have reject at Final Production --
reject is at Design/UX)
32      15. Log Start Production Run          [LogWorkflowHandler -
auto-advance]          (=Major Step 29)
33      16. Replicate to Preview
[ReplicateToAgentHandler -> push-to-preview]          (=Major Step 30)
34      17. Cache Flush Preview
[ReplicateToAgentHandler -> flush-preview]          (=Major Step 31)
35      18. Check Delayed Release Date
[CheckDelayedReleaseDateHandler - auto-advance]          (=Major Step 32)
36      19. Report Delayed Release as Comment
[SetDelayedReleaseAsCommentHandler - auto-advance] (=Major Step 33)
37      20. Send Pending Publication Email
[EmailNotificationsHandler - auto-advance]          (=Major Step 34)
38      **[DIFFERS]**
Notification group: lt-wf-simple-notification
39      Template:
.../simple/pending-publication/en.txt
40      21. Wait to Publish          [Dynamic Participant -
> AbsoluteTimeAutoAdvancer]          (=Major Step 35)
41      22. OR Split: Force Deploy / Cancel
(=Major Step 36)
42      +-- Branch 1: Force Deploy -> continue
43      +-- Branch 2: Cancel -> Jump to Cancel Processing
44      23. Replicate to Production
[ReplicateToAgentHandler -> push-to-production]          (=Major Step 37)
45      24. Cache Flush Production
[ReplicateToAgentHandler -> flush-production]          (=Major Step 38)
46      25. Send Finished Publication Email
[EmailNotificationsHandler - auto-advance]          (=Major Step 39)
47      **[DIFFERS]**
Notification group: lt-wf-simple-notification
48      Template:
.../simple/finished-publication/en.txt
49      26. Jump to End (skip cancel processing)          [JumpToNodeHandler]
(=Major Step 40)
50      27. Cancel Workflow Processing          [LogWorkflowHandler -
auto-advance]          (=Major Step 41)
28. Inform Content Author about Cancellation
[EmailNotificationsHandler - auto-advance]          (=Major Step 42)

```

51	Template: ../simple/cancelled/en.txt	**[DIFFERS]**
52	29. Log End of Workflow auto-advance] (=Major Step 43)	[LogWorkflowHandler -
53		

2.3 Key Differences Between Variants

Difference	Major Review	Simple Review
Approval gates	3 (Design/UX, Editorial, Final Production)	1 (Final Production only)
Reject-to-rework location	At Design/UX Review (Step 16, Branch 2)	At Final Production Review (Step 14, Branch 3)
Final Production OR Split	2 branches (Approve / Cancel)	3 branches (Approve / Cancel / Reject)
First-loop skip target	"Hand over to the Design / UX Review"	"Start the Final Production Review"
Set steppedBack=true step	Present (Step 10) after rework OR Split	Absent -- no explicit steppedBack=true between rework and Final Production
Email template path	/etc/.../lifetech/major/...	/etc/.../lifetech/simple/...
Notification group (pending/finished)	lt-wf-major-notification	lt-wf-simple-notification
Total steps	43	29
Total email notifications	8	5
Design/UX Review	Steps 10-16	Not present
Editorial Review	Steps 17-22	Not present

2.5 Participants and Roles

Role	AEM Group	Major Review	Simple Review
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Content Author / Initiator	lt-wf-author	Step 3 (Select Web Ops), Step 8 (Rework)	Step 3 (Select Web Ops), Step 7 (Rework)
Design / UX Reviewer	lt-wf-design-reviewer	Step 15	Not used
Editorial Reviewer	lt-wf-editorial-reviewer	Step 21	Not used
Final Production Reviewer (Web Ops)	Dynamic (PropertiesParticipantChooser)	Step 27	Step 13
Rejection Notification Recipients	lt-wf-rejection-notification	Step 7 (email)	Step 6 (email)
Major Notification Recipients	lt-wf-major-notification	Steps 34, 39 (email)	Not used
Simple Notification Recipients	lt-wf-simple-notification	Not used	Steps 20, 25 (email)

3. Future Workflow Definitions (AEM Cloud Service + EDS)

3.1 Architecture Approach

Both workflows are migrated preserving the same business logic, participants, and transitions. Changes are technical (infrastructure-level) only and **apply equally to both variants** since they share the same handler codebase. Refactoring a handler once applies to both workflows.

3.2 Major Review -- Future Step Sequence (38 steps)

```

1  1. Log Start Workflow [LogWorkflowHandler - refactored]
2  2. Set Rework Parameters (steppedBack=false) [PlaceParametersHandler - refactored]
3  3. Select Web Operations Team [Dialog Participant Step -> lt-wf-author]
4  4. Place Web Ops Team on Workflow Metadata [DialogParameterHandler - refactored]
5  5. Jump Over Rework if First Loop [JumpToNodeIfSetHandler - refactored]

```

```

6 6. Log Rework Content [LogWorkflowHandler -
  refactored]
7 7. Send Rework Notification Email
  [EmailNotificationsHandler - refactored]
8 8. Rework Content Step [Dynamic Participant -
  > Java InitiatorParticipantChooser]
9 <-- CHANGED: ECMA ->
  Java ParticipantStepChooser (if ECMA script not supported in cloud)
10 9. OR Split: Done with Rework / Cancel
11   --- Branch 1: Log "Done" -> continue
12   --- Branch 2: Log "Cancel" -> Goto Cancel Processing
13 <-- CHANGED: OOTB
  Goto Step replaces JumpToNodeHandler
14 10. Set steppedBack=true
  [PlaceParametersHandler - refactored]
15 11. Report Metadata [LogWorkflowHandler -
  refactored]
16 12. Publish to Preview Tier [Sling Content
  Distribution API]
17 <-- CHANGED: Replaces
  agent-based ReplicateToAgentHandler
18 (Cache Flush step REMOVED -- EDS CDN handles cache invalidation
  automatically)
19 13. Notify Design Reviewer Email
  [EmailNotificationsHandler - refactored]
20 14. Review: UX / Design [Participant Step ->
  lt-wf-design-reviewer]
21 15. OR Split: Approve / Reject / Cancel
22   --- Branch 1: Log "Approve" -> continue
23   --- Branch 2: Log "Reject" -> Goto step 6 (Rework) <--
  CHANGED: OOTB Goto Step
24   --- Branch 3: Log "Cancel" -> Goto Cancel Processing <--
  CHANGED: OOTB Goto Step
25 16. Log Start Editorial Review [LogWorkflowHandler -
  refactored]
26 17. Publish to Preview Tier [Sling Content
  Distribution API] <-- CHANGED
27 (Cache Flush REMOVED)
28 18. Notify Editorial Reviewer Email
  [EmailNotificationsHandler - refactored]
29 19. Review: Editorial [Participant Step ->
  lt-wf-editorial-reviewer]
30 20. OR Split: Approve / Cancel
31   --- Branch 1: Log "Approve" -> continue
32   --- Branch 2: Log "Cancel" -> Goto Cancel Processing <--
  CHANGED: OOTB Goto Step
33 21. Log Start Final Production Review [LogWorkflowHandler -
  refactored]
34 22. Publish to Preview Tier [Sling Content
  Distribution API] <-- CHANGED
35 (Cache Flush REMOVED)
36 23. Notify Final Production Reviewer Email
  [EmailNotificationsHandler - refactored]
37 24. Review: Final Production [Dynamic Participant -
  > PropertiesParticipantChooser]
38 (Dialog: Absolute Timer Delay -- NO CHANGE)
39 25. OR Split: Approve / Cancel
40   --- Branch 1: Approve + Place Delayed Release Date -> continue
41   --- Branch 2: Cancel -> Goto Cancel Processing <--
  CHANGED: OOTB Goto Step
42 26. Log Start Production Run [LogWorkflowHandler -
  refactored]
43 27. Publish to Preview Tier [Sling Content
  Distribution API] <-- CHANGED
44 (Cache Flush REMOVED)
45 28. Check Delayed Release Date
  [CheckDelayedReleaseDateHandler - refactored]

```



```

46 29. Report Delayed Release as Comment
   [SetDelayedReleaseAsCommentHandler - refactored]
47 30. Send Pending Publication Email
   [EmailNotificationsHandler - refactored]
48 31. Waiting to Publish                                [Dynamic Participant -
   > AbsoluteTimeAutoAdvancer]
49                                     (NO CHANGE -- OOTB,
   backed by Sling Jobs)
50 32. OR Split: Force Deploy / Cancel
51     +-- Branch 1: Log "Force Deploy" -> continue
52     +-- Branch 2: Log "Cancel" -> Goto Cancel Processing    <--
   CHANGED: OOTB Goto Step
53 33. Publish to Production                                [Sling Content
   Distribution API]    <-- CHANGED
54     (Cache Flush REMOVED -- EDS CDN invalidates automatically)
55 34. Send Finished Publication Email
   [EmailNotificationsHandler - refactored]
56 35. Goto End (skip cancel processing)                    [OOTB Goto Step]
   <-- CHANGED
57 36. Cancel Workflow Processing                            [LogWorkflowHandler -
   refactored]
58 37. Inform Content Author about Cancellation
   [EmailNotificationsHandler - refactored]
59 38. Log End of Workflow                                  [LogWorkflowHandler -
   refactored]
60

```

3.3 Simple Review -- Future Step Sequence (25 steps)

All changes below are the **same type of changes** as Major Review (same handlers refactored, same OOTB replacements).

```

1  1. Log Start Workflow                                [LogWorkflowHandler -
   refactored]
2  2. Set Rework Parameters (steppedBack=false)
   [PlaceParametersHandler - refactored]
3  3. Select Web Operations Team                        [Dialog Participant
   Step -> lt-wf-author]
4  4. Place Web Ops Team on Workflow Metadata
   [DialogParameterHandler - refactored]
5  5. Jump Over Rework if First Loop
   [JumpToNodeIfSetHandler - refactored]
6  6. Rework Content (Email Notification)
   [EmailNotificationsHandler - refactored]
7  7. Rework Content Step                                [Dynamic Participant -
   > Java InitiatorParticipantChooser]
8                                     <-- CHANGED: ECMA ->
   Java ParticipantStepChooser (If ECMA script is not supported in Cloud)
9  8. OR Split: Done / Cancel
10     +-- Branch 1: Log "Done" -> continue
11     +-- Branch 2: Log "Cancel" -> Goto Cancel Processing    <--
   CHANGED: OOTB Goto Step
12  9. Log Start Final Production Review                  [LogWorkflowHandler -
   refactored]
13 10. Publish to Preview Tier                            [Sling Content
   Distribution API]    <-- CHANGED
14     (Cache Flush REMOVED)
15 11. Notify Final Production Reviewer
   [EmailNotificationsHandler - refactored]
16 12. Review: Final Production                          [Dynamic Participant -
   > PropertiesParticipantChooser]
17     (Dialog: Absolute Timer Delay -- NO CHANGE)
18 13. OR Split: Approve / Cancel / Reject
19     +-- Branch 1: Approve + Place Delayed Release Date -> continue

```

```

20    +-- Branch 2: Cancel -> Goto Cancel Processing          <--
    CHANGED: OOTB Goto Step
21    +-- Branch 3: Reject -> Goto Rework Content            <--
    CHANGED: OOTB Goto Step
22    14. Log Start Production Run                          [LogWorkflowHandler -
    refactored]
23    15. Publish to Preview Tier                          [Sling Content
    Distribution API]          <-- CHANGED
24        (Cache Flush REMOVED)
25    16. Check Delayed Release Date                        [CheckDelayedReleaseDateHandler - refactored]
26    17. Report Delayed Release as Comment                 [SetDelayedReleaseAsCommentHandler - refactored]
27    18. Send Pending Publication Email                    [EmailNotificationsHandler - refactored]
28    19. Wait to Publish                                  [Dynamic Participant -
    > AbsoluteTimeAutoAdvancer]
29                                                    (NO CHANGE -- OOTB,
    backed by Sling Jobs)
30    20. OR Split: Force Deploy / Cancel
31        +-- Branch 1: Force Deploy -> continue
32        +-- Branch 2: Cancel -> Goto Cancel Processing      <--
    CHANGED: OOTB Goto Step
33    21. Publish to Production                            [Sling Content
    Distribution API]          <-- CHANGED
34        (Cache Flush REMOVED -- EDS CDN invalidates automatically)
35    22. Send Finished Publication Email                    [EmailNotificationsHandler - refactored]
36    23. Goto End (skip cancel processing)                 [OOTB Goto Step]
    <-- CHANGED
37    24. Cancel Workflow Processing + Cancellation Email
38    25. Log End of Workflow                              [LogWorkflowHandler -
    refactored]
39

```

4. Business Functionality Assessment

4.1 Functionality Preservation Status (Both Variants)

Capability	Current Behavior	Future Behavior	Major	Simple
Multi-stage approval	Sequential gates with OR Splits	Same	Fully Preserved	Fully Preserved
Rework loop (reject → rework → re-review)	Workflow jumps back to rework step	Same (OOTB Goto Step)	Fully Preserved	Fully Preserved
First-loop rework skip	Custom handler skips rework on	Refactored handler, same logic	Fully Preserved	Fully Preserved

	initial submission			
Web Ops team selection	Author selects from dialog dropdown	Same dialog	Fully Preserved	Fully Preserved
Delayed/scheduled release	Reviewer sets date via datepicker	Same mechanism (AbsoluteTimeAutoAdvancer)	Fully Preserved	Fully Preserved
Force deploy	OR Split allows immediate publication	Same OR Split	Fully Preserved	Fully Preserved
Email notifications	Templated emails to participants	Same handler, refactored	Fully Preserved	Fully Preserved
Cancel from any stage	Jump to shared cancel handler	OOTB Goto Step to cancel handler	Fully Preserved	Fully Preserved
Comment history in emails	Assembled from workflow history	Same logic	Fully Preserved	Fully Preserved
Participant roles and groups	AEM user groups	Same groups	Fully Preserved	Fully Preserved
Preview before review	Replicated to preview agent	Published to EDS Preview(.aem.page)	Fully Preserved	Fully Preserved

Production publish	Replicated to production agent	Published via Cloud Distribution - > EDS CDN	Fully Preserved	Fully Preserved
Inbox delayed release visibility	Comment shows release date	Same mechanism	Fully Preserved	Fully Preserved

4.2 Impact by Role

Role	What Changes	What Stays the Same	Applies To
Content Author	Nothing	Initiates workflow, selects Web Ops, reworks on rejection	Both
Design/UX Reviewer	Preview URL changes to EDS (.aem.page)	Reviews, approves/rejects /cancels from Inbox	Major only
Editorial Reviewer	Preview URL changes to EDS (.aem.page)	Reviews, approves/cancels from Inbox	Major only
Web Ops (Final Production)	Preview URL changes to EDS (.aem.page)	Sets release date, approves/cancels/forces deploy	Both
All Participants	Email links point to Cloud author + EDS preview URLs	Notifications, comment history, Inbox experience	Both

4.3 What Is NOT Changing (Both Variants)

- Workflow initiation process
 - Number and sequence of approval gates (per variant)
 - Participant assignment logic
 - OR Split branching options
 - Rework loop behavior
 - Delayed release date selection and timer
 - Force deploy option
 - Email notification content and recipients
 - Workflow cancellation flow
 - AEM Inbox experience
 - Variant selection (authors still choose between Major and Simple Review when initiating)
-

5. Technical Change Details

All technical changes documented below apply equally to both the Major Review and Simple Review workflows, as they share the same custom handler codebase.

5.1 Steps Removed (No Replacement Needed)

Step	Major Instances	Simple Instances	Current Behavior	Why Removed
Cache Flush Preview	4	2	Replicates to flush-preview agent	EDS CDN handles cache invalidation automatically. No Dispatcher in EDS.
Cache Flush Production	1	1	Replicates to flush-production agent	Same -- EDS CDN push-based invalidation.

Total steps removed: 5 (Major), 3 (Simple)

5.2 Steps Replaced with OOTB AEM Cloud Features

Current Step	Major Instances	Simple Instances	Replacement	Why
Replicate to Preview	4	2	Sling Content Distribution API	Agent-based replication not available in Cloud. Reference: AEM Cloud Replication
Replicate to Production	1	1	Sling Content Distribution API	Same -- content flows through xWalk pipeline to EDS CDN
JumpToNodeHandler (unconditional jumps)	8	4	OOTB Goto Step (com.day.cq.workflow.impl.process.GotoStep)	<code>createTransition()</code> modifies workflow model at runtime -- incompatible with Cloud's immutable infrastructure

5.3 Steps Refactored (Same Behavior, Updated Code)

5.3.1 LogWorkflowHandler

Instances: Major: 14, Simple: 8

Changes: Felix SCR → OSGi DS annotations; remove `AbstractResourceWorkflowProcess` base class.

Business impact: None.

5.3.2 PlaceParametersHandler

Instances: Major: 2, Simple: 1

Note: Major Review has `PlaceParametersHandler` setting `steppedBack=true` after the rework OR Split (Step 10). Simple Review **does not have this step** -- see Open Item 8.5.

Changes: Felix SCR → OSGi DS; fix deprecated `new Long()` → `Long.valueOf()`.

Business impact: None.

5.3.3 DialogParameterHandler

Instances: Major: 2, Simple: 2

Changes: Felix SCR → OSGi DS annotations.

Business impact: None.

5.3.4 JumpToNodeIfSetHandler

Instances: Major: 1, Simple: 1

What it does: Checks `steppedBack` property to skip rework on first loop. Jump target differs per variant:

- Major: targets "Hand over to the Design / UX Review"
- Simple: targets "Start the Final Production Review"

Changes: Implement `WorkflowProcess` directly; remove `createTransition()` and `SimpleRoute`; use pre-defined routes; Felix SCR → OSGi DS. Add pre-defined transition in both workflow models from this step to its respective jump target.

Business impact: None -- same conditional logic in both variants.

5.3.5 EmailNotificationsHandler

Instances: Major: 8, Simple: 5

Why it cannot be used as-is:

Issue	Detail	Severity
<code>com.day.cq.mailer</code> <code>.MailService</code> removed	AEM Cloud Service has no direct SMTP access	Critical
Felix SCR annotations	Must migrate to OSGi DS	High

JCR Session as instance variable	Unsafe in Cloud -- instances restart frequently	High
Email templates at /etc/	Deprecated content path	Medium

Changes:

1. Replace MailService with cloud-compatible email service .
2. Migrate to OSGi DS annotations
3. Remove Session instance variable -- use method-local ResourceResolver
4. Relocate email templates from /etc/ to /conf/
5. Make host URLs configurable via OSGi

Email templates requiring relocation:

Template	Major Review Path	Simple Review Path
Rework rejection	.../major/review/rejected/en.txt	.../simple/review/rejected/en.txt
Design review notification	.../major/review/design/en.txt	N/A (not used)
Editorial review notification	.../major/review/editorial/en.txt	N/A (not used)
Final production notification	.../major/review/webops/en.txt	.../simple/review/webops/en.txt
Pending publication	.../major/pending-publication/en.txt	.../simple/pending-publication/en.txt
Finished publication	.../major/finished-publication/en.txt	.../simple/finished-publication/en.txt

Cancellation	.../major/cancelled/en.txt	.../simple/cancelled/en.txt
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Total templates to relocate: 7 (Major) + 5 (Simple) = 12 templates

Business impact: None -- same emails, same recipients, same content.

5.3.6 CheckDelayedReleaseDateHandler

Instances: 1 in each variant.

Changes: Felix SCR → OSGi DS; `SimpleDateFormat` → `DateTimeFormatter` (thread-safety).

Business impact: None.

5.3.7 SetDelayedReleaseAsCommentHandler

Instances: 1 in each variant.

Changes: Felix SCR → OSGi DS; `SimpleDateFormat` → `DateTimeFormatter`; review `updateTransFlag()` static call.

Business impact: None.

6. Clarifications Needed

6.1 Delayed Release Default Behavior

Current behavior: When no activation date is entered by the Final Production reviewer, `CheckDelayedReleaseDateHandler` applies a default and sets the release time to NOW + 1 day (configurable via `delayOffset` parameter). This means a page is **never published immediately** after final approval -- there is always at least a 1-day delay unless someone explicitly uses the "Force the Deployment" path.

Clarification needed from business:

Is the 1-day default delay a deliberate business requirement (e.g., to provide a review window before content goes live, or to coordinate with regional publishing schedules), or is it a technical safeguard that may no longer be needed?

Recommendation: If there is no specific business requirement to delay page activation, the recommendation is to publish immediately upon approval. The delayed release mechanism would remain fully available for cases where a reviewer explicitly sets a future activation date.